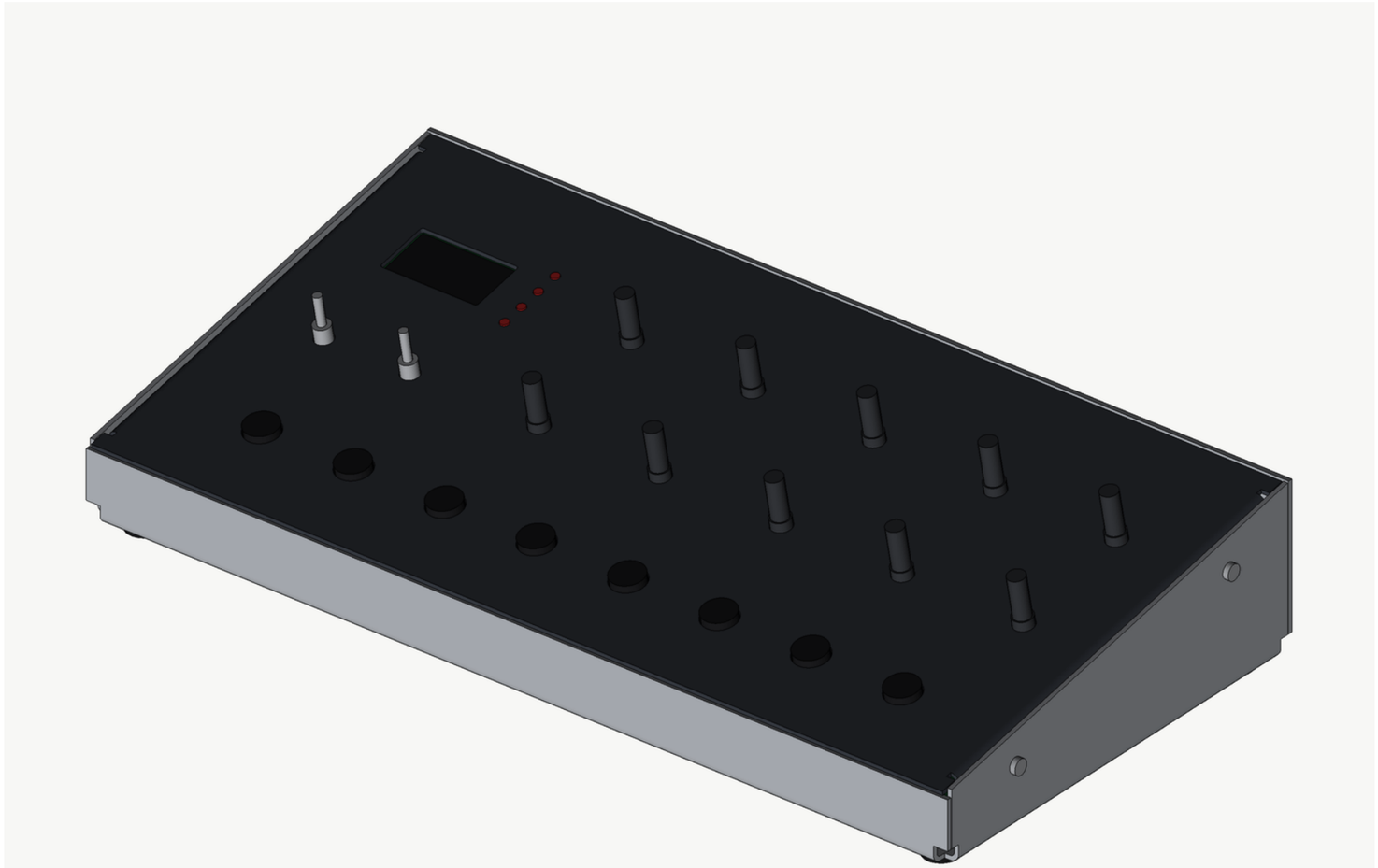


Checks: 39 pass, 0 warn, 0 fail. All 39 are on the last sheet.

Every check passes.

Sizes

Size	mm
Overall, everything	324 × 160 × 71.4
bottom_shell, formed	320 × 160 × 58
bottom_shell, flat	429.54 × 236.21
top_panel, formed	316.2 × 160.91 × 18
top_panel, flat	346.54 × 160.91
Panel slope	11.31°

Parts list

#	Component	Qty	Source
1	bottom_shell	1	sheet metal
2	top_panel	1	sheet metal
3	screw_M3x6_pan	4	stand-in
4	rubber_foot_12x4	4	stand-in
5	control_board_pcb	1	STEP: control_board.step
6	oled_096	1	STEP: control_board.step
7	led_3mm	4	STEP: control_board.step
8	pot_9mm	10	STEP: control_board.step
9	toggle_m6	2	STEP: control_board.step
10	button_12mm	8	STEP: control_board.step
11	PEM CLS-M3-1 nut	4	clinched in panel lips

Parameters

Parameter	Value
W	320
D	160
Hf	26
Hb	58
rule.material	Aluminium 5052-H32 (placeholder)
rule.thickness	1.6
rule.bend_radius	1.6
rule.k_factor	0.42
corner_gap	0.4
panel_clearance	0.3
front_lip_clearance	0.5
relief_depth	None
relief_width	2
corner_relief	2
lip_depth	18
pem_edge_cl	4.8
board_gap	10
board_inset	8

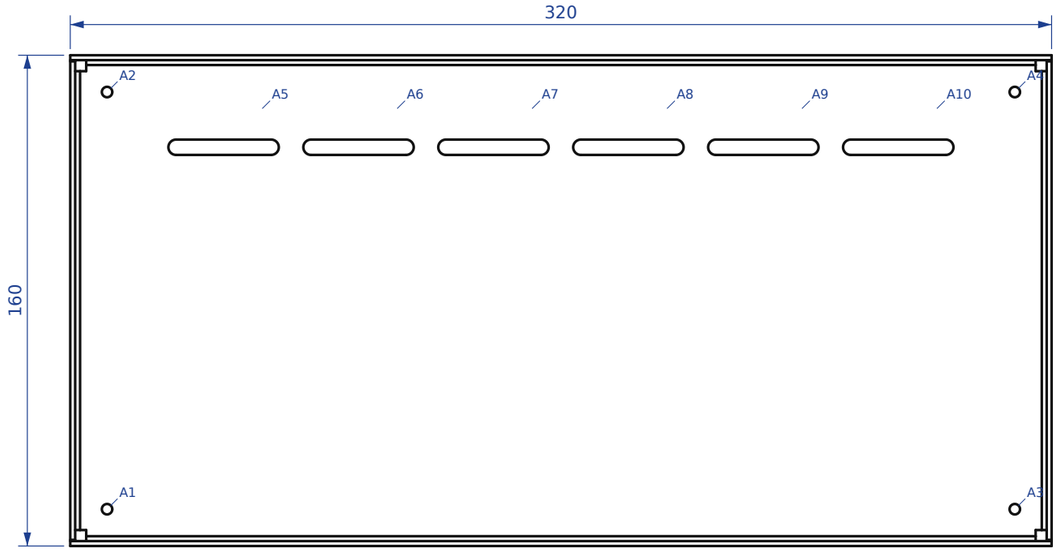
NOTES

1. Inside bend radius R1.6, K-factor 0.42.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Material, bend radius and K-factor are PLACEHOLDERS: use your fab's values.
5. Hole sizes for parts are typical values; confirm with datasheets.

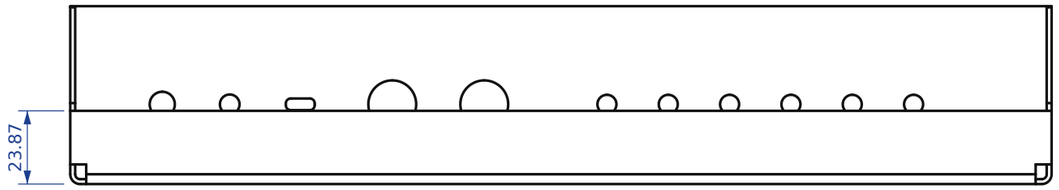
Synth case: assembly

qfold

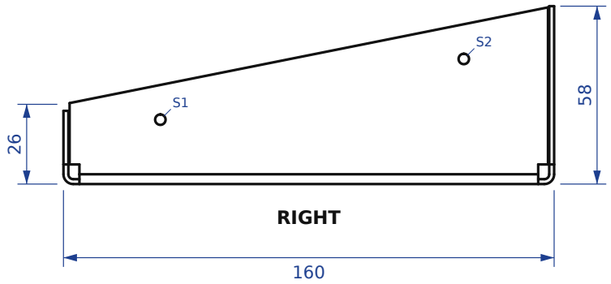
PART	SCALE	SHEET
ASSEMBLY	NTS	1 of 6
MATERIAL	UNITS	REV / DATE
Aluminium 5052-H32 (placeholder), t=1.6	mm	A 2026-11-09



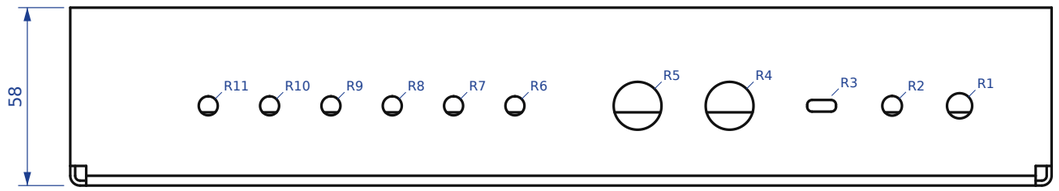
TOP



FRONT



RIGHT



REAR (viewed from behind)

Base holes (top view, from front-left outer corner)

Tag	Feature	X	Y	Size
A1	foot screw M3	12	12	Ø3.4
A2	foot screw M3	12	148	Ø3.4
A3	foot screw M3	308	12	Ø3.4
A4	foot screw M3	308	148	Ø3.4
A5	vent slot	50	130	36×5 R2.5
A6	vent slot	94	130	36×5 R2.5
A7	vent slot	138	130	36×5 R2.5
A8	vent slot	182	130	36×5 R2.5
A9	vent slot	226	130	36×5 R2.5
A10	vent slot	270	130	36×5 R2.5

Rear wall holes (from rear view lower-left)

Tag	Feature	X	Z	Size
R1	DC barrel jack	290	26	Ø8.2
R2	power switch (M6)	268	26	Ø6.4
R3	USB-C	245	26	9.4×3.8 R1.6
R4	MIDI DIN in	215	26	Ø15.6
R5	MIDI DIN out	185	26	Ø15.6
R6	3.5 mm jack	145	26	Ø6.2
R7	3.5 mm jack	125	26	Ø6.2
R8	3.5 mm jack	105	26	Ø6.2
R9	3.5 mm jack	85	26	Ø6.2
R10	3.5 mm jack	65	26	Ø6.2
R11	3.5 mm jack	45	26	Ø6.2

Cheek holes (from front, base)

Tag	Feature	Y	Z	Size
S1	M3 clearance (right)	31.59	21	Ø3.4
S2	M3 clearance (right)	130.55	40.79	Ø3.4
S3	M3 clearance (left)	31.59	21	Ø3.4
S4	M3 clearance (left)	130.55	40.79	Ø3.4

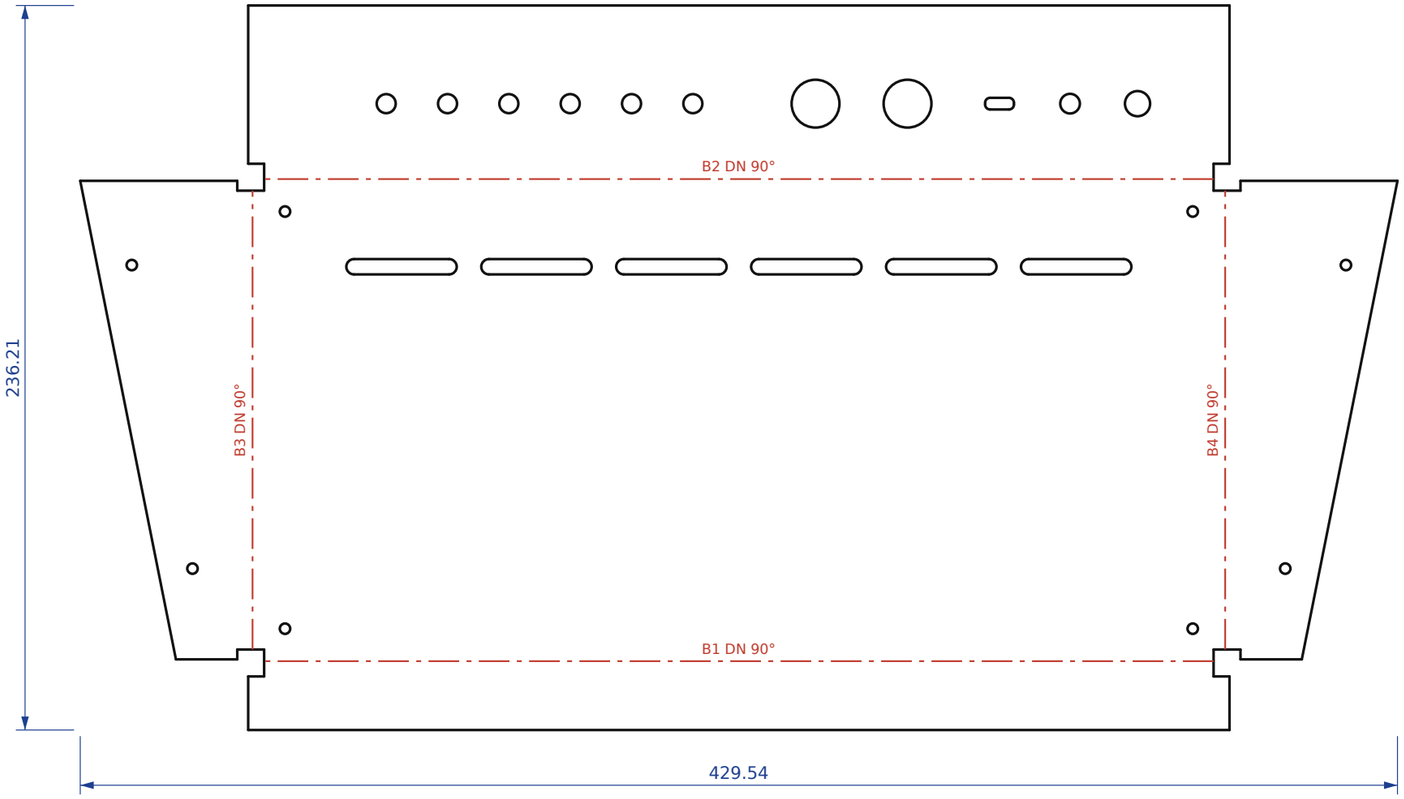
NOTES

1. Inside bend radius R1.6, K-factor 0.42.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Material, bend radius and K-factor are PLACEHOLDERS: use your fab's values.
5. Hole sizes for parts are typical values; confirm with datasheets.
6. Third-angle projection. Hole positions from outer faces.

Bottom shell: views

qfold

PART	SCALE	SHEET
BOTTOM SHELL	1:2	2 of 6
MATERIAL	**UNITS**	**REV / DATE**
Aluminium 5052-H32 (placeholder), t=1.6	mm	A 2026-11-09



FLAT PATTERN 429.54 × 236.21

Bends

Tag	Bend	Dir	Angle	In. R	Length	Allow.	Centre line
B1	front	DN	90°	R1.6	309.6	3.57	Y 22.45
B2	back	DN	90°	R1.6	309.6	3.57	Y 179.62
B3	right	DN	90°	R1.6	149.6	3.57	X 56.18
B4	left	DN	90°	R1.6	149.6	3.57	X 373.35

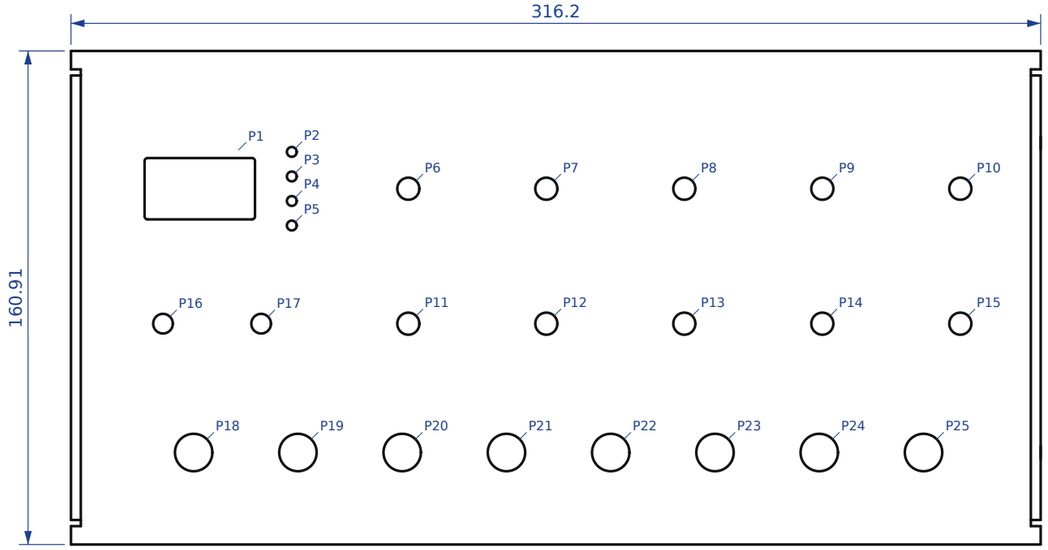
NOTES

1. Inside bend radius R1.6, K-factor 0.42.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Material, bend radius and K-factor are PLACEHOLDERS: use your fab's values.
5. Hole sizes for parts are typical values; confirm with datasheets.
6. Corner reliefs and closed corners are in the DXF.
7. Outside face shown. DN = bend away from viewer. DXF is the cutting master.

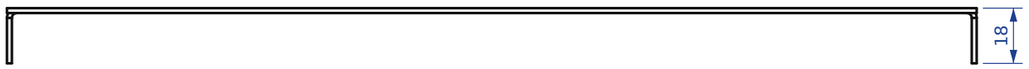
Bottom shell: flat pattern

qfold

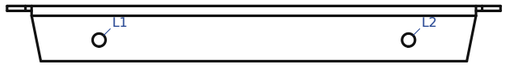
PART	SCALE	SHEET
BOTTOM SHELL	1:2	3 of 6
MATERIAL	UNITS	REV / DATE
Aluminium 5052-H32 (placeholder), t=1.6	mm	A 2026-11-09



TOP (outside face)



FRONT EDGE



RIGHT

FLAT PATTERN 346.54 × 160.91

Panel holes (from front-left corner)

Tag	Feature	X	Y	Size
P1	OLED window	42	116	36×20 R1
P2	3 mm LED	72	128	Ø3.2
P3	3 mm LED	72	120	Ø3.2
P4	3 mm LED	72	112	Ø3.2
P5	3 mm LED	72	104	Ø3.2
P6	9 mm pot (M7 bushing)	110	116	Ø7.2
P7	9 mm pot (M7 bushing)	155	116	Ø7.2
P8	9 mm pot (M7 bushing)	200	116	Ø7.2
P9	9 mm pot (M7 bushing)	245	116	Ø7.2
P10	9 mm pot (M7 bushing)	290	116	Ø7.2
P11	9 mm pot (M7 bushing)	110	72	Ø7.2
P12	9 mm pot (M7 bushing)	155	72	Ø7.2
P13	9 mm pot (M7 bushing)	200	72	Ø7.2
P14	9 mm pot (M7 bushing)	245	72	Ø7.2
P15	9 mm pot (M7 bushing)	290	72	Ø7.2
P16	toggle switch (M6)	30	72	Ø6.4
P17	toggle switch (M6)	62	72	Ø6.4
P18	12 mm button	40	30	Ø12.2
P19	12 mm button	74	30	Ø12.2
P20	12 mm button	108	30	Ø12.2
P21	12 mm button	142	30	Ø12.2
P22	12 mm button	176	30	Ø12.2
P23	12 mm button	210	30	Ø12.2
P24	12 mm button	244	30	Ø12.2
P25	12 mm button	278	30	Ø12.2

Lip holes (Z below top face)

Tag	Feature	Y	Z↓	Size
L1	PEM CLS-M3 nut	30	11.1	Ø4.22
L2	PEM CLS-M3 nut	130.91	11.1	Ø4.22
L3	PEM CLS-M3 nut	30	11.1	Ø4.22
L4	PEM CLS-M3 nut	130.91	11.1	Ø4.22

Bends

Tag	Bend	Dir	Angle	In. R	Length
B1	panel_right	DN	90°	R1.6	144.91
B2	panel_left	DN	90°	R1.6	144.91

NOTES

1. Inside bend radius R1.6, K-factor 0.42.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Material, bend radius and K-factor are PLACEHOLDERS: use your fab's values.
5. Hole sizes for parts are typical values; confirm with datasheets.
6. Third-angle projection. Press PEM nuts from inside face.

Top panel

qfold

PART	SCALE	SHEET
TOP PANEL	1:2	4 of 6
MATERIAL	UNITS	REV / DATE
Aluminium 5052-H32 (placeholder), t=1.6	mm	A 2026-11-09

Part	Tag	Feature	Size	X	Y	Z	Face	Facing
bottom_shell	A1	foot screw M3	Ø3.4	12	12	0	base	−Z down
bottom_shell	A2	foot screw M3	Ø3.4	12	148	0	base	−Z down
bottom_shell	A3	foot screw M3	Ø3.4	308	12	0	base	−Z down
bottom_shell	A4	foot screw M3	Ø3.4	308	148	0	base	−Z down
bottom_shell	A5	vent slot	36×5 R2.5	50	130	0	base	−Z down
bottom_shell	A6	vent slot	36×5 R2.5	94	130	0	base	−Z down
bottom_shell	A7	vent slot	36×5 R2.5	138	130	0	base	−Z down
bottom_shell	A8	vent slot	36×5 R2.5	182	130	0	base	−Z down
bottom_shell	A9	vent slot	36×5 R2.5	226	130	0	base	−Z down
bottom_shell	A10	vent slot	36×5 R2.5	270	130	0	base	−Z down
bottom_shell	R1	DC barrel jack	Ø8.2	30	160	26	back_wall	+Y back
bottom_shell	R2	power switch (M6)	Ø6.4	52	160	26	back_wall	+Y back
bottom_shell	R3	USB-C	9.4×3.8 R1.6	75	160	26	back_wall	+Y back
bottom_shell	R4	MIDI DIN in	Ø15.6	105	160	26	back_wall	+Y back
bottom_shell	R5	MIDI DIN out	Ø15.6	135	160	26	back_wall	+Y back
bottom_shell	R6	3.5 mm jack	Ø6.2	175	160	26	back_wall	+Y back
bottom_shell	R7	3.5 mm jack	Ø6.2	195	160	26	back_wall	+Y back
bottom_shell	R8	3.5 mm jack	Ø6.2	215	160	26	back_wall	+Y back
bottom_shell	R9	3.5 mm jack	Ø6.2	235	160	26	back_wall	+Y back
bottom_shell	R10	3.5 mm jack	Ø6.2	255	160	26	back_wall	+Y back
bottom_shell	R11	3.5 mm jack	Ø6.2	275	160	26	back_wall	+Y back
bottom_shell	S1	M3 clearance (right)	Ø3.4	320	31.59	21	cheek_right	+X right
bottom_shell	S2	M3 clearance (right)	Ø3.4	320	130.55	40.79	cheek_right	+X right
bottom_shell	S3	M3 clearance (left)	Ø3.4	0	31.59	21	cheek_left	−X left
bottom_shell	S4	M3 clearance (left)	Ø3.4	0	130.55	40.79	cheek_left	−X left
top_panel	P1	OLED window	36×20 R1	43.9	113.75	48.75	panel	0.00, -0.20, 0.98
top_panel	P2	3 mm LED	Ø3.2	73.9	125.51	51.1	panel	0.00, -0.20, 0.98
top_panel	P3	3 mm LED	Ø3.2	73.9	117.67	49.53	panel	0.00, -0.20, 0.98
top_panel	P4	3 mm LED	Ø3.2	73.9	109.83	47.97	panel	0.00, -0.20, 0.98
top_panel	P5	3 mm LED	Ø3.2	73.9	101.98	46.4	panel	0.00, -0.20, 0.98
top_panel	P6	9 mm pot (M7 bushing)	Ø7.2	111.9	113.75	48.75	panel	0.00, -0.20, 0.98
top_panel	P7	9 mm pot (M7 bushing)	Ø7.2	156.9	113.75	48.75	panel	0.00, -0.20, 0.98
top_panel	P8	9 mm pot (M7 bushing)	Ø7.2	201.9	113.75	48.75	panel	0.00, -0.20, 0.98
top_panel	P9	9 mm pot (M7 bushing)	Ø7.2	246.9	113.75	48.75	panel	0.00, -0.20, 0.98
top_panel	P10	9 mm pot (M7 bushing)	Ø7.2	291.9	113.75	48.75	panel	0.00, -0.20, 0.98
top_panel	P11	9 mm pot (M7 bushing)	Ø7.2	111.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P12	9 mm pot (M7 bushing)	Ø7.2	156.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P13	9 mm pot (M7 bushing)	Ø7.2	201.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P14	9 mm pot (M7 bushing)	Ø7.2	246.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P15	9 mm pot (M7 bushing)	Ø7.2	291.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P16	toggle switch (M6)	Ø6.4	31.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P17	toggle switch (M6)	Ø6.4	63.9	70.6	40.12	panel	0.00, -0.20, 0.98
top_panel	P18	12 mm button	Ø12.2	41.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P19	12 mm button	Ø12.2	75.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P20	12 mm button	Ø12.2	109.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P21	12 mm button	Ø12.2	143.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P22	12 mm button	Ø12.2	177.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P23	12 mm button	Ø12.2	211.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P24	12 mm button	Ø12.2	245.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	P25	12 mm button	Ø12.2	279.9	29.42	31.88	panel	0.00, -0.20, 0.98
top_panel	L1	PEM CLS-M3 nut	Ø4.22	318.1	31.59	21	lip_right	+X right
top_panel	L2	PEM CLS-M3 nut	Ø4.22	318.1	130.55	40.79	lip_right	+X right
top_panel	L3	PEM CLS-M3 nut	Ø4.22	1.9	31.59	21	lip_left	−X left
top_panel	L4	PEM CLS-M3 nut	Ø4.22	1.9	130.55	40.79	lip_left	−X left

NOTES

1. Hole centres on the outside surface, in assembly coordinates.
2. Facing: the direction the hole opens to, out of the part.

Holes as assembled

qfold

PART	SCALE	SHEET
SYNTH CASE	NTS	5 of 6
MATERIAL	UNITS	REV / DATE
Aluminium 5052-H32 (placeholder), t=1.6	mm	A 2026-11-09

Checks: 39 pass, 0 warn, 0 fail

Check	Result	
bottom_shell: valid solid	valid=True, solids=1, faces=96	PASS
bottom_shell: no self-intersection	folded fuse loses 2.62e-10 mm³, flat fuse loses -5.82e-11 mm³ (limit 1.4e-04)	PASS
bottom_shell: flat pattern matches fold	folded−flat volume 295.448442 mm³, bend allowance predicts 295.448442	PASS
bottom_shell: hole to bend	tightest 7.10 mm (base foot screw M3 to bend front), need ≥ 4.80	PASS
bottom_shell: hole to edge	closest 7.10 mm (base foot screw M3), need ≥ 1.60	PASS
bottom_shell: web between holes	narrowest 8.00 mm (base vent slot / vent slot), need ≥ 1.60	PASS
bottom_shell: holes inside their face	every hole lies within its face	PASS
bottom_shell: minimum hole size	all holes ≥ 1.60 mm	PASS
bottom_shell: bend corner relief	bends end ≥ 2.83 mm apart, need ≥ 1.60	PASS
bottom_shell: flange length	shortest flat flange 20.67 mm (bend front), need ≥ 6.40	PASS
top_panel: valid solid	valid=True, solids=1, faces=70	PASS
top_panel: no self-intersection	folded fuse loses 0.00e+00 mm³, flat fuse loses -7.28e-11 mm³ (limit 8.5e-05)	PASS
top_panel: flat pattern matches fold	folded−flat volume 93.235471 mm³, bend allowance predicts 93.235471	PASS
top_panel: hole to bend	tightest 5.79 mm (lip_right PEM CLS-M3 nut to bend panel_right), need ≥ 4.80	PASS
top_panel: hole to edge	closest 4.79 mm (lip_right PEM CLS-M3 nut), need ≥ 1.60	PASS
top_panel: web between holes	narrowest 4.80 mm (panel 3 mm LED / 3 mm LED), need ≥ 1.60	PASS
top_panel: holes inside their face	every hole lies within its face	PASS
top_panel: minimum hole size	all holes ≥ 1.60 mm	PASS
top_panel: hardware edge distance	4 hardware holes meet their centre-to-edge minimum	PASS
top_panel: bend corner relief	bends end ≥ 309.80 mm apart, need ≥ 1.60	PASS
top_panel: flange length	shortest flat flange 14.80 mm (bend panel_right), need ≥ 6.40	PASS
bottom_shell.step reads back	1 solid(s), volume 137537.324 mm³ (model 137537.324)	PASS
top_panel.step reads back	1 solid(s), volume 84810.012 mm³ (model 84810.012)	PASS
assembly.step reads back	36 solid(s), volume 335258.135 mm³ (model 335258.135)	PASS
bottom_shell vs top_panel	overlap 0.000000 mm³, closest 0.300 mm, need ≥ 0.2	PASS
screw heads on cheeks	4 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
feet on base	4 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
board parts against panel	20 touching, widest gap 0.0000 mm (≤ 0.01), 6 clear of it, no overlap	PASS
control board vs bottom_shell	overlap 0.000000 mm³, closest 5.870 mm, need ≥ 2	PASS
faces of one part keep their gaps	7 unjoined face pairs, tightest 0.400 mm (bottom_shell: back_wall vs cheek_right), need ≥ 0.2	PASS
no part overlaps another	11 top-level parts, none overlapping	PASS
screws on cheek and lip holes	4 parts, worst 0.000 mm across, 0.000 mm along, 0.000°	PASS
feet on their base holes	4 parts, worst 0.000 mm across, 0.000 mm along, 0.000°	PASS
board parts on their panel holes	25 parts, worst 0.000 mm across, 0.000 mm along, 0.000°	PASS
every panel hole has a part, every part a hole	25 panel holes, 25 board parts	PASS
bottom_shell: fits sheet	flat 429.5 × 236.2 mm, sheet limit 1200 × 600	PASS
bottom_shell_flat.dxf reads back	18 circles, 44 lines, 16 arcs, 4 bends, cut 2178.6 mm, extent 429.538 × 236.206	PASS
top_panel: fits sheet	flat 346.5 × 160.9 mm, sheet limit 1200 × 600	PASS
top_panel_flat.dxf reads back	28 circles, 28 lines, 4 arcs, 2 bends, cut 1806.3 mm, extent 346.538 × 160.911	PASS

NOTES

1. Run on every build; also in report.json.
2. Limits come from the project's rules (a shop's, or qfold's defaults in dfm.py).

Checks		qfold
PART **SYNTH CASE**	SCALE NTS	SHEET 6 of 6
MATERIAL Aluminium 5052-H32 (placeholder), t=1.6	UNITS mm	REV / DATE A 2026-11-09